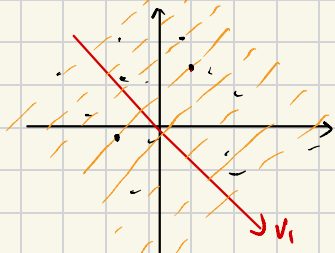


## Lecture 11 non linear dimensionality deduction

PCA

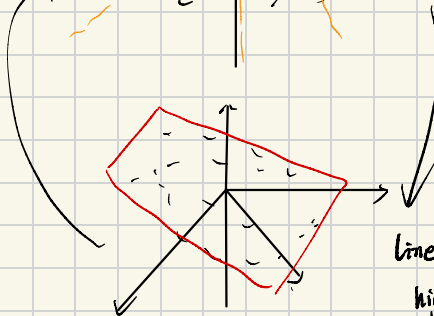


iso contours (the points have the same PCA co-ef)

non-linear



apply feature transform  $\Phi$  (kernel trick)



linear component in higher-dim space.

kernel PCA

matrix

Data:  $\{x_1, \dots, x_n\}$

$$X = [x_1, \dots, x_n]$$

transform:  $x_i \rightarrow \phi(x_i) = \phi_i$

$$\Phi = [\phi_1, \dots, \phi_n]$$

Assume: the data is centered in high-dim f.s.  $\sum_i \phi_i = 0$

1) co-variance matrix

$$C = \frac{1}{n} \sum_i \phi(x_i) \phi(x_i)^T$$

2) Find eigen values / vectors:  $(\lambda_j, v_j)$ :

$$C v_j = \lambda_j v_j$$

$$\frac{1}{n} \sum_i \phi(x_i) \phi(x_i)^T \cdot v_j = \lambda_j v_j$$

scalars

Then  $v_j$  has the form  $v_j = \sum_{i=1}^n a_{ij} \phi(x_i)$  (linear combo of  $\phi(x_i)$ )

$$v_j = \Phi a_j, \quad a_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{nj} \end{bmatrix}$$

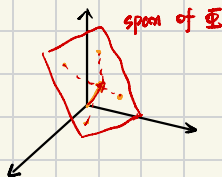
It suffices to compute  $a_j$  to get  $v_j$

$$\frac{1}{n} \sum_i \phi(x_i) \phi(x_i)^T \Phi a_j = \lambda_j \Phi a_j$$

$\Phi \Phi^T$

$$\frac{1}{n} \Phi \Phi^T \Phi a_j = \lambda_j \Phi a_j$$

K eigen-vector



## Lecture 11 non linear dimensionality deduction

$$\frac{1}{n} \Phi^T \Phi \Phi^T a_j = \lambda_j \Phi^T \Phi a_j$$

$$\frac{1}{n} K \cdot K \cdot a_j = \lambda_j K \cdot a_j$$

1) if  $K$  is invertible:  $\Rightarrow K a_j = \lambda_j a_j$

2) if  $K$  is not invertible: there are  $\lambda_j = 0$ , which do not affect PCA

suppose  $(a_j, \lambda_j)$  are eigenpairs of  $K$

$$K a_j = n \lambda_j a_j, \quad \lambda_j \neq 0, \quad a_j^T a_j = 1$$

$$K K a_j = n \lambda_j K a_j$$

$$n \lambda_j K a_j = n^2 \lambda_j^2 a_j$$

$$n^2 \lambda_j^2 a_j = n^2 \lambda_j^2 a_j$$

PC should be normalized:  $v_i^T \cdot v_j = 1$

$$a_j^T \Phi^T \Phi a_j = 1$$

$$a_j^T K \cdot a_j = 1$$

$$n \lambda_j a_j^T a_j = 1$$

$$\Rightarrow a_j = \frac{1}{\sqrt{n \lambda_j}} a_j$$

kernel centering:

$$\tilde{\phi}(x_i) = \phi(x_i) - \frac{1}{n} \sum_j \phi(x_j) \quad \text{subtract mean of } \phi_i$$

$$= \phi(x_i) - \frac{1}{n} \Phi \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \phi(x_i) - \frac{1}{n} \Phi \mathbf{1}$$

vector of ones

$$\tilde{\Phi} = [\tilde{\phi}(x_1) \dots \tilde{\phi}(x_n)] = \Phi - \left[ \frac{1}{n} \Phi \mathbf{1}_n \dots \right]$$

$$= \Phi - \frac{1}{n} \Phi \mathbf{1} \mathbf{1}^T$$

$$= \Phi \left( \mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right)$$

$$\tilde{K} = \tilde{\Phi}^T \tilde{\Phi} = \underbrace{\left( \mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right)}_K \Phi^T \Phi \underbrace{\left( \mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right)}$$

Note: the original problem is a  $d$ -dim eigen problem  
 $\Rightarrow$  kernelized  $n$ -dim problem

summary of KPCA: given  $X = [x_1 \dots x_n]$

1) calculate kernel matrix  $K = [k(x_i, x_j)]_{ij}$

2) center kernel:  $\tilde{K} = \left( \mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) K \left( \mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right)$

3) Find the top  $D$  eigenvectors:  $\tilde{K} a_j = \lambda_j a_j, j = 1 \dots D$

4) scale:  $a_j \leftarrow \frac{1}{\sqrt{\lambda_j}} a_j$

5) project:  $x_*: z_j = \tilde{\Phi}(x_*)^T \tilde{\Phi} a_j = \underbrace{\tilde{K}_*^T}_{\tilde{T}} a_j, j = 1 \dots D$

centered test vector

## Lecture 11 pre image problem

Given PCA coefficient  $\mathbf{z}$ , reconstruct  $\hat{\mathbf{x}}$

$$\hat{\mathbf{x}} = \mathbf{V}\mathbf{z} = \sum_j \mathbf{v}_j z_j$$

For KPCA

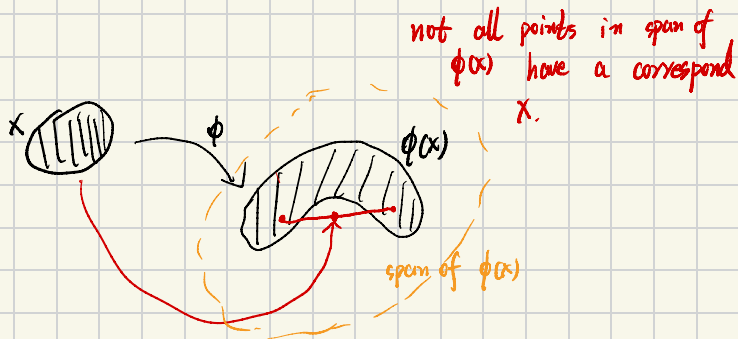
Given  $\mathbf{a}$   $\mathbf{z}$ ,  $\hat{\phi} = \sum_j \mathbf{v}_j z_j = \sum_j \Phi \mathbf{a}_j z_j = \Phi (\sum_j \mathbf{a}_j z_j)$

let  $\phi(\mathbf{x}) = \begin{bmatrix} x^2 \\ x \end{bmatrix}$

$x_1=1 \Rightarrow \phi(\mathbf{x}_1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$x_2=2 \Rightarrow \phi(\mathbf{x}_2) = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$

$\hat{\phi} = \phi(\mathbf{x}_1) + \phi(\mathbf{x}_2) = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$



General problem:

Given  $\hat{\phi} = \Phi \alpha = \sum_j \alpha_j \phi(\mathbf{x}_j)$

minimize norm in  $\mathcal{H}$

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\phi(\mathbf{x}) - \hat{\phi}\|^2$$

$$= \arg \min_{\mathbf{x}} K(\mathbf{x}, \mathbf{x}) - 2 \sum_j \alpha_j K(\mathbf{x}, \mathbf{x}_j) + \underbrace{\sum_j \alpha_j \alpha_i K(\mathbf{x}_j, \mathbf{x}_i)}_{\text{constant}}$$

$$= \arg \min_{\mathbf{x}} K(\mathbf{x}, \mathbf{x}) - 2 \sum_j \alpha_j K(\mathbf{x}, \mathbf{x}_j)$$

Solution

1) select  $\mathbf{x} \in X$  (training data),  $\mathbf{x}_i$  w/  $\phi(\mathbf{x}_i)$  closest to  $\hat{\phi}$

$$\arg \min_j K(\mathbf{x}_j, \mathbf{x}_j) - 2 \mathbf{K}_j^T \alpha$$

2) solve optim numerically using an optimizer.

3) Suppose  $K(\mathbf{x}, \mathbf{x}) = 1$ ,  $K(\mathbf{x}_i, \mathbf{x}_j) \geq 0$  (e.g. Gaussian)

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} - \sum_i \alpha_i K(\mathbf{x}, \mathbf{x}_i) = \arg \max_{\mathbf{x}} \sum_i \alpha_i K(\mathbf{x}, \mathbf{x}_i)$$

$$= \arg \max_{\mathbf{x}} \left( \sum_i \alpha_i K(\mathbf{x}, \mathbf{x}_i) \right)^2$$

$\Rightarrow$  iterative algom:  $\hat{\mathbf{x}} \leftarrow \frac{\sum_i \alpha_i K'(\|\mathbf{x}_i - \hat{\mathbf{x}}\|^2) \cdot \mathbf{x}_i}{\sum_i \alpha_i K'(\|\mathbf{x}_i - \hat{\mathbf{x}}\|^2)}$

$K(\|\mathbf{x}_i - \mathbf{x}_j\|^2) = \frac{1}{q^{\|\mathbf{x}_i - \mathbf{x}_j\|^2}}$  Gaussian